

BANKVISION

Credit Analytics Dashboard Project Report

Prepared by: Sidharth Khare | Tool: Microsoft Power BI | Data: Kaggle Credit Risk Dataset

1. Executive Summary

This report presents a comprehensive credit risk analysis of a bank loan portfolio containing 31,677 loan records, built in Power BI from a public Kaggle credit risk dataset. The analysis covers portfolio composition, loan performance by risk grade, customer demographics, and actionable lending recommendations.

31,677

TOTAL LOANS

\$306M

TOTAL LOAN AMOUNT

78.45%

GOOD LOAN %

21.55%

DEFAULT RATE

Key Findings

- **Risk Alert:** 1 in 5 loans defaults (21.55%), well above a healthy portfolio benchmark.
- **Grade G Criticality:** Grade G borrowers show a near-certain default rate of 98.44%.
- **Demographics:** The portfolio is dominated by young borrowers (avg. age 27.73), an elevated risk segment.
- **Debt Burden:** Average Debt-to-Income (DTI) across the portfolio is 16.96%.
- **Concentration:** Grades A and B together make up roughly two-thirds of all loans (10,369 + 10,183 of 31,677).

2. Portfolio Overview

The portfolio splits 78.45% fully paid to 21.55% defaulted. Education, Medical, and Venture are the three largest loan purposes both by volume and by amount funded.

24,852	6,825	11.04%	\$9.66K
FULLY PAID	DEFAULTED	AVG INTEREST RATE	AVG LOAN AMOUNT

Loans by Purpose (volume)

Purpose	Loan Count	Loan Amount
Education	6.3K	\$60M
Medical	5.9K	\$55M
Venture	5.6K	\$54M
Personal	5.4K	—
Debt Consolidation	5.1K	—
Home Improvement	3.5K	—

Amount figures shown only where labeled on the source dashboard; remaining purposes were not individually labeled.

Loan Grade Distribution

A	10,369
B	10,183
C	6,318
D	3,555
E	952
F	236
G	64

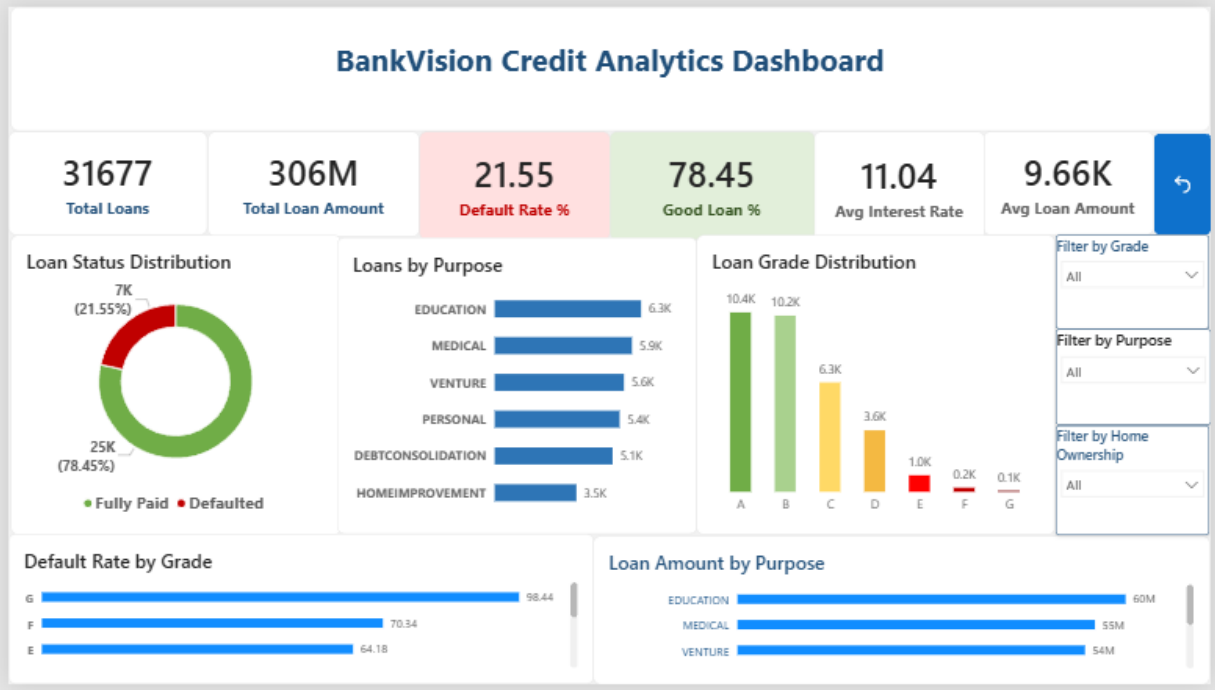


Figure 1 — BankVision Credit Analytics Dashboard: Overview page

3. Loan Performance & Risk Analysis

6,825

BAD LOANS

\$75M

BAD LOAN AMOUNT

1,252

HIGH RISK LOANS

3.95%

HIGH RISK %

Grade-wise Portfolio Summary

Grade	Total Loans	Default Rate	Loan Amount	Bad Loans
A	10,369	9.56%	\$89,185,025	991
B	10,183	15.93%	\$102,583,625	1,622
C	6,318	20.31%	\$58,631,625	1,283
D	3,555	58.76%	\$38,691,550	2,089
E	952	64.18%	\$12,317,225	611
F	236	70.34%	\$3,491,875	166
G	64	98.44%	\$1,100,525	63
Total	31,677	21.55%	\$306,001,450	6,825

Default Rate by Loan Purpose (highest to lowest)

Debt Consolidation > Medical > Home Improvement > Personal > Education > Venture. Debt Consolidation and Medical loans carry the highest default risk; Education and Venture loans carry the lowest.



Figure 2 — Loan Performance & Risk Analysis page

4. Customer Insights & Profile

27.73

AVG. CUSTOMER AGE

\$66.49K

AVG. ANNUAL INCOME

16.96%

AVG. DEBT-TO-INCOME

Home Ownership Profile

Ownership	Share of Loans	Default Rate
Rent	50.73% (16K)	35.1%
Mortgage	41.32% (13K)	12.5%
Own	7.61% (2K)	6.9%
Other	< 1%	30.8%

Renters default at roughly 3–5x the rate of homeowners — home ownership status is a strong risk signal.

Income Segment & Credit History

- **Income mix:** Mid (\$30–60K) 13.6K loans, High (\$60–100K) 10.2K, Premium (\$100K+) 4.5K, Low (<\$30K) 2.4K.
- **Prior default history:** 82.23% of borrowers (26K) have no previous default on file; 17.77% (6K) do.

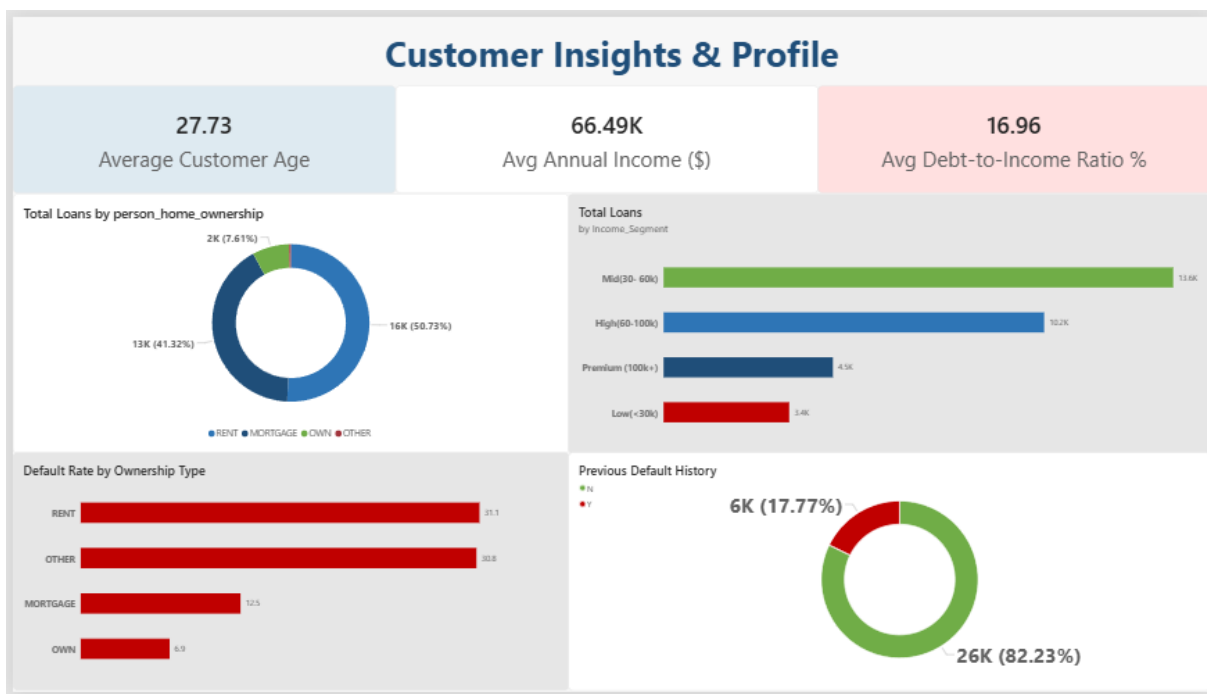


Figure 3 — Customer Insights & Profile page

5. Key Insights & Recommendations

Portfolio Health Scorecard

31,677 TOTAL LOANS	21.55% DEFAULT RATE	\$306M TOTAL LOAN AMOUNT	11.04 AVG INTEREST RATE
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Default rate sits at 21.55% against a 15% target ceiling — 6.55 points over threshold.

CRITICAL	Grade G borrowers: 98.44% default rate. Action: stop lending to Grade G immediately.
OPPORTUNITY	Grade A/B borrowers: 9.56%–15.93% default, well below the 21.55% portfolio average. Action: expand lending to Grade A/B.
ACTION NEEDED	Average DTI: 16.96%. Action: set a hard cap at 35% DTI for all new loan applications.

Grade-wise Action Plan

Grade	Risk Level	Action	Priority
A / B	Low	Increase lending	Low
C	Moderate	Maintain current standards	Medium
D / E / F / G	Critical	Stop lending now	Critical

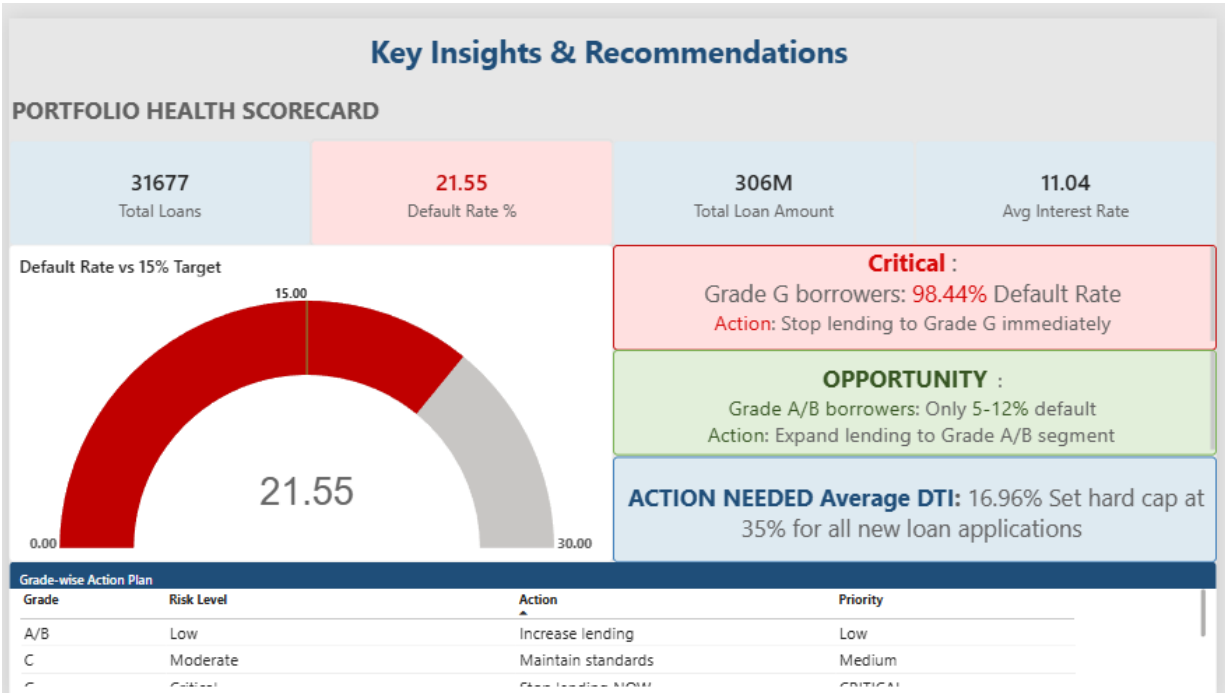


Figure 4 — Key Insights & Recommendations page

6. Data & Methodology

Source: Credit Risk Dataset by laotse, Kaggle (CC0: Public Domain).
kaggle.com/datasets/laotse/credit-risk-dataset

The original dataset (32,581 records) was cleaned in Power Query by removing age (>80) and employment length (>60 years) anomalies, reducing the working dataset to 31,677 records. Missing interest rate values were imputed using the column mean (11.04%).

Technical Implementation (DAX)

Total Loan Amount = SUM(loan_amnt)

Default Rate % = DIVIDE([Bad Loans], [Total Loans])

High Risk % = DIVIDE([High Risk Loans], [Total Loans])

Tools: Microsoft Power BI, Power Query, DAX.